

QUESTION BANK FOR UT-02

UNIT- 3

2 Marks

- 1) Compare 3G and 4G Mobile Generations on the basis of
Data speed
Technology

Ans:

Parameter	3G	4G
Data speed	The maximum upload rate of 3G technology is 5 Mbps The maximum download rate of 3G technology is 21 Mbps	The maximum upload rate of 4G technology is 500 Mbps • The maximum download rate of 3G technology is 1 Gbps
Technology Used	The technology used in 3G is WCDMA (Wideband Code Division Multiple Access), Digital Broadband Packet Data CDMA 2000, UMTS, EDGE, etc.	The technology used in 4G is LTE (Long-Term Evolution), and WiMAX (Worldwide Interoperability for Microwave Access).

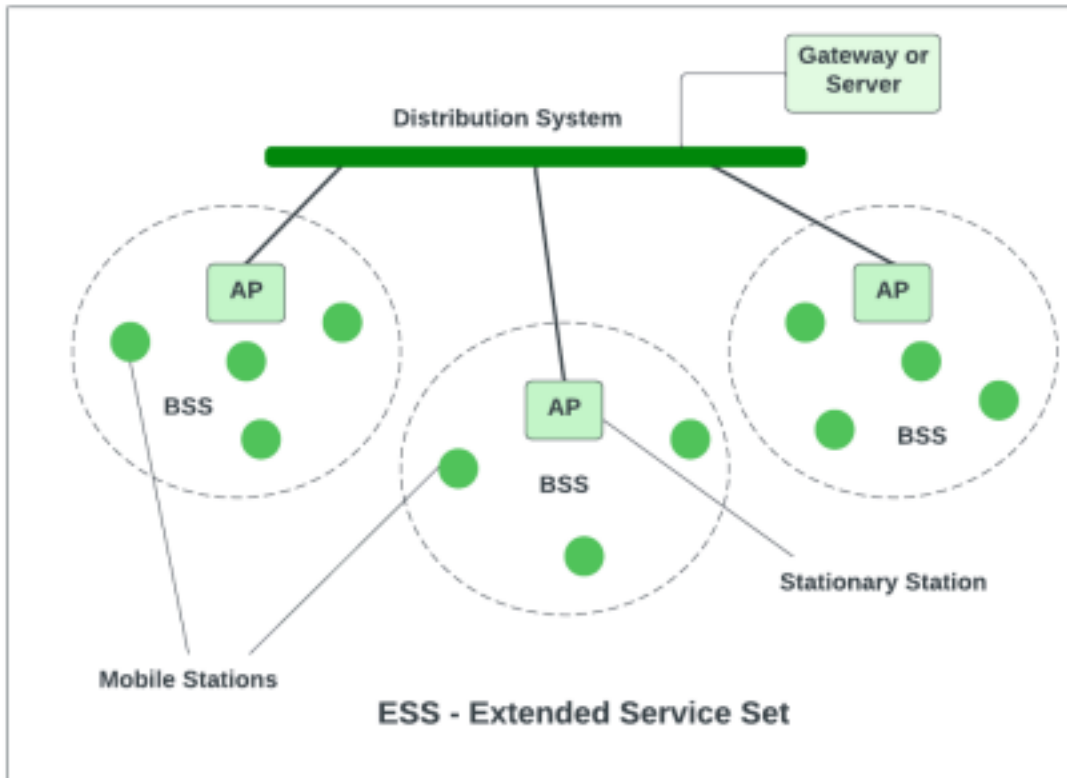
- 2) List IEEE 802.11 standards for network Ans:

802.11 – Original wireless LAN standard 802.11a – 5 GHz, 54 Mbps
802.11b – 2.4 GHz, 11 Mbps
802.11g – 2.4 GHz, 54 Mbps
802.11p – 5.9 GHz

4 Marks

1) Explain 802.11 architecture

Ans:



- IEEE 802.11 standard, popularly known as **WiFi**, lays down the architecture and specifications of **wireless LANs (WLANs)**.
- WiFi or WLAN uses high-frequency radio waves instead of cables for connecting the devices in LAN.
- Users connected by WLANs can move around within the area of network coverage.
- **Stations (STA)** Stations comprises of all devices and equipment that are connected to the wireless LAN.
- **Basic Service Set (BSS)** A basic service set is a group of stations communicating at the **physical layer level**.
- **Extended Service Set (ESS)** It is a set of all connected BSS.
- **Distribution System (DS)** It connects access points in ESS.

2) Explain working of CRC with example

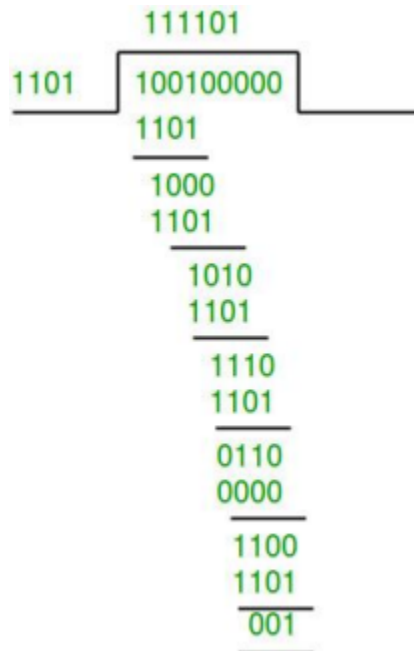
Ans:

example:

Data = 100100, Generator Polynomial (Key) = $x^3 + x^2 + 1$ (1101)

Append Zeros: Append 3 zeros to the data: 100100000

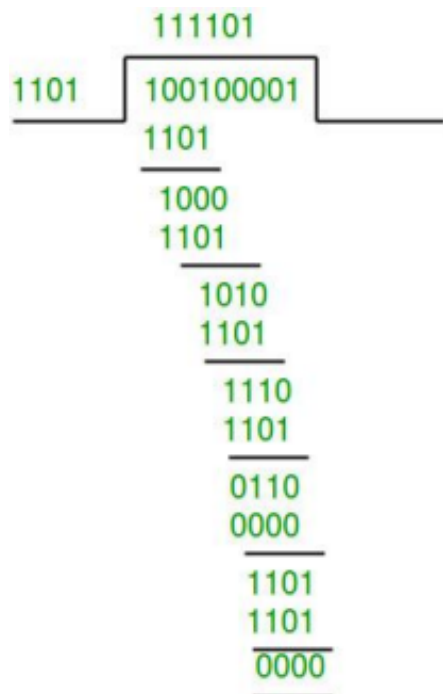
Sender Side



The remainder is 001. Thus the data sent is 100100001.

Receiver Side

Code word received at the receiver side 100100001



Steps in CRC Working

1. Dataword

The sender first takes the original data bits called the **dataword**.

Example:

Dataword = **100100**

2. Generator Polynomial

A predefined binary number called the **generator polynomial (divisor)** is selected.

Example:

Generator = $x^3 + x^2 + 1$ (**1101**)

3. Append Zeros

The sender adds **(n – 1) zeros** to the end of the dataword, where **n is the number of bits in the generator**.

Generator length = 4

Zeros added = 3

New data:

100100000

4. Perform Binary Division

The modified data is divided by the generator using **modulo-2 division (XOR operation)**.

This division produces a **remainder**.

Example remainder:

001

5. Append CRC Bits

The remainder obtained from the division is called the **CRC bits**. These bits are appended to the original data.

Transmitted data:

100100001

6. Checking at Receiver

The receiver divides the received data by the **same generator polynomial**. • If the **remainder is 0**, the data is **accepted as correct**.

- If the **remainder is not 0**, it means **an error occurred during transmission**.

UNIT - 4

2 Marks

1) List any four Application layer protocol

Ans: FTP(File transfer protocol)

SMTP(simple mail transfer protocol)

HTTP(hypertext transfer protocol)

DHCP(dynamic host configuration protocol)

2) Define Data Encapsulation and Subnet

Ans:

Data Encapsulation -Encapsulation is the process of adding additional information when data is traveling in an OSI or TCP/IP model.

Subnet -A Subnet (Subnetwork) is a smaller logical network created by dividing a larger network into multiple smaller networks.

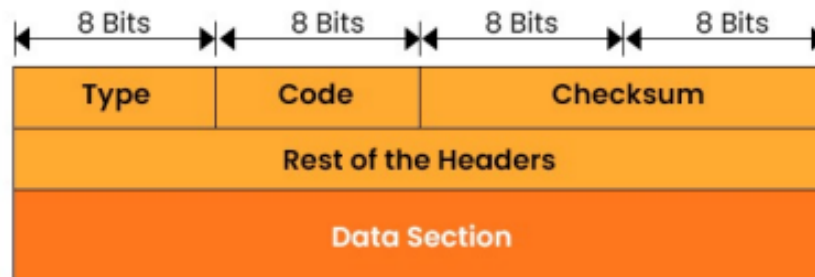
3) List any two IP functions

Ans:Logical Addressing

Routing of data packets between networks

4) Draw ICMP message format

Ans:



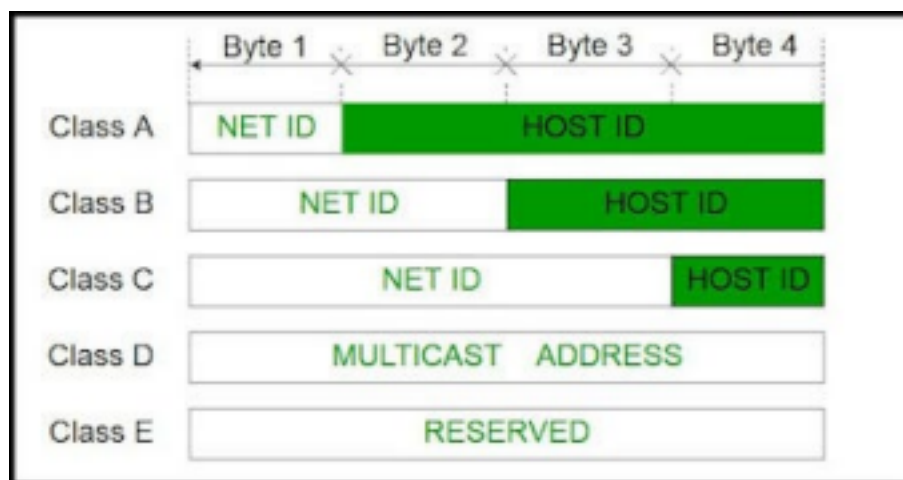
ICMP Message Format

1) Compare IPV4 and IPV6

Ans:

IPv4	IPv6
Uses a 32-bit IP address.	Uses a 128-bit IP address.
Uses decimal dot-separated notation (e.g., 192.168.0.1).	Uses hexadecimal colon-separated notation (e.g., 2001:db8::1).
Provides a limited address space of about 4.3 billion addresses.	Provides an extremely large address space for future growth.
Uses address classes (A, B, C, D, and E).	Does not use address classes.
Example address: 66.94.29.13.	Example address: 2001:db8::1.

2) Describe various IP address classes with suitable example Ans:



1. Class A

First Octet Range: 0 – 127

Default Subnet Mask: 255.0.0.0

Network and Host: 8 bits for network, 24 bits for host

Used for: Very large networks

Example:

IP Address: **10.0.0.1**

2. Class B

First Octet Range: 128 – 191

Default Subnet Mask: 255.255.0.0

Network and Host: 16 bits for network, 16 bits for host

Used for: Medium-sized networks

Example:

IP Address: **172.16.5.4**

3. Class C

First Octet Range: 192 – 223

Default Subnet Mask: 255.255.255.0

Network and Host: 24 bits for network, 8 bits for host

Used for: Small networks

Example:

IP Address: **192.168.1.10**

4. Class D

First Octet Range: 224 – 239

Used for: Multicasting (sending data to multiple devices)

Example:

IP Address: **230.1.1.1**

5. Class E

First Octet Range: 240 – 255

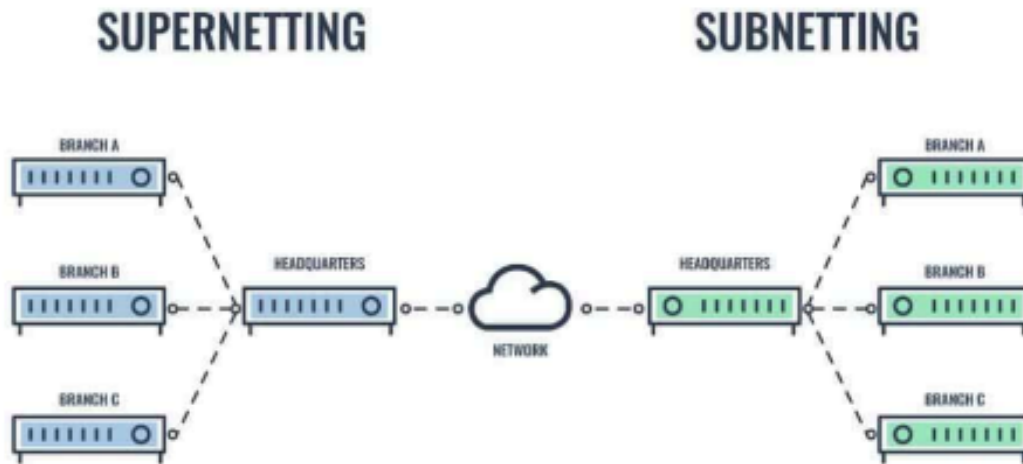
Used for: Experimental and research purposes

Example:

IP Address: **250.1.1.1**

3) Explain Subnetting and Supernetting

Ans:



Subnetting and supernetting are techniques used to manage IP address allocation and improve network organization.

Subnetting

- Subnetting divides a larger network into smaller, more manageable subnetworks (subnets).
- This is done by borrowing bits from the host portion of the IP address to create additional network bits.
- For example, a network with the address 192.168.1.0/24 could be subnetted into smaller networks, such as 192.168.1.0/26, 192.168.1.64/26, etc., making it easier to control traffic and allocate IP addresses efficiently.

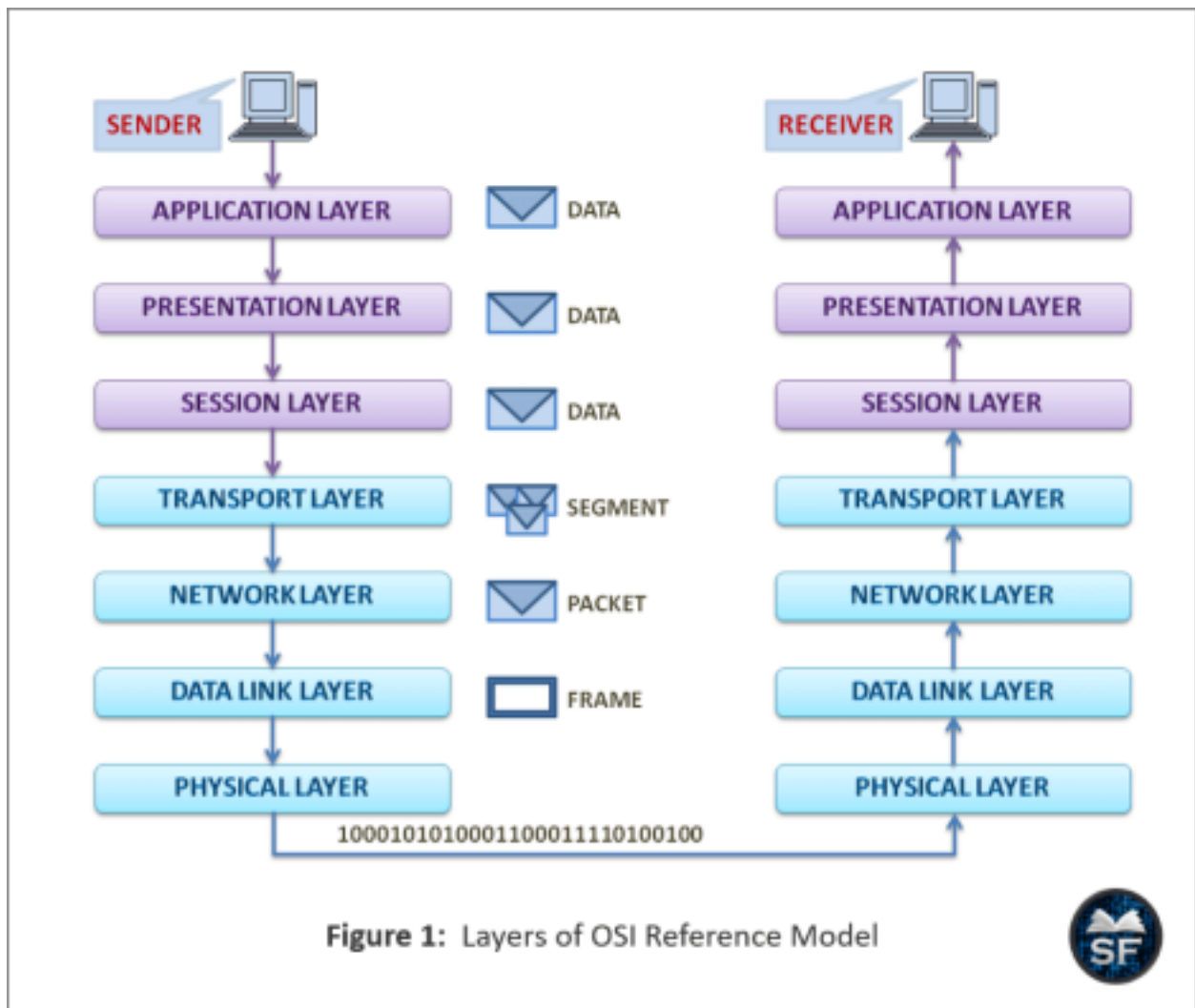
Supernetting

- Supernetting, on the other hand, combines multiple networks into a larger network by borrowing bits from the network portion of the address.
- This results in a single, broader address range.
- Supernetting is often used by Internet Service Providers (ISPs) to aggregate IP addresses, simplifying routing tables and improving scalability.
- For instance, the networks 192.168.0.0/24 and 192.168.1.0/24 can be supernetted into 192.168.0.0/23.

6 Marks

1) Explain OSI reference model with suitable diagram Ans:

- The OSI (Open Systems Interconnection) Model is a set of rules that explains how different computer systems communicate over a network.
- OSI Model was developed by the International Organization for Standardization (ISO).
- The OSI Model consists of 7 layers and each layer has specific functions and responsibilities.
- This layered approach makes it easier for different devices and technologies to work together.



The OSI model is divided into seven layers, as follows –

- Physical Layer – Its function is to transmit individual bits from one node to another over a physical medium.
- Data Link Layer – It is responsible for the reliable transfer of data frames from one node to another connected by the physical layer.
- Network Layer – It manages the delivery of individual data packets from source to destination through appropriate addressing and routing.
- Transport Layer – It is responsible for delivery of the entire message from the source host to destination host.
- Session Layer – It establishes sessions between users and offers services like dialog control and synchronization.
- Presentation Layer – It monitors syntax and semantics of transmitted information through translation, compression, and encryption.
- Application Layer – It provides high-level APIs (application program interface) to the users.

2) Explain HTTP, SMTP and ARP protocol in detail

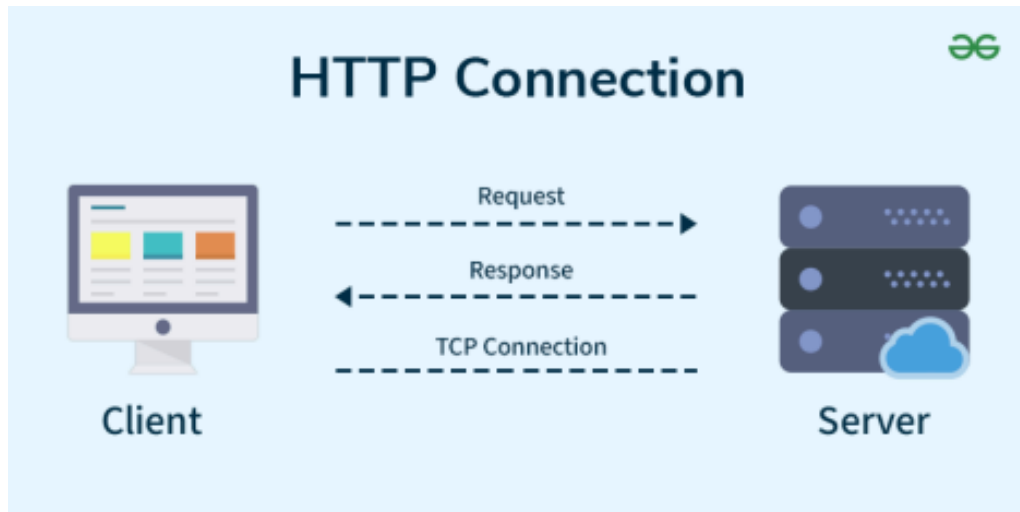
Ans:

HTTP

HTTP (Hyper Text Transfer Protocol) is an application-layer protocol used to transfer hypertext documents over the World Wide Web.

It defines how web browsers (clients) and web servers communicate to request and deliver web content.

- Used for web page communication.
- Works on a client–server model.
- Transfers text, images, audio, video, and other multimedia content.
- Not secure by default, unless used with HTTPS.



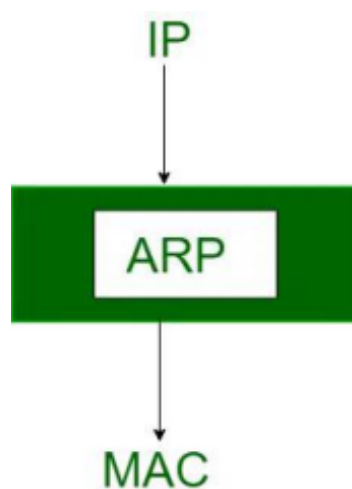
SMTP

SMTP (Simple Mail Transfer Protocol) is an application-layer protocol used for sending and routing outgoing emails over a network.

It is responsible for transferring email messages from the sender's mail server to the receiver's mail server in a reliable manner.

- Used only for sending emails, not for receiving them.
- Transfers emails from the sender's mail server to the receiver's mail server.
- Removes the email from the outgoing queue after successful delivery.

ARP



ARP (Address Resolution Protocol) maps an IPv4 address to the corresponding MAC address on a local network so frames can be delivered to the correct device.

- Resolves IP addresses to MAC addresses for local communication
- Works only within the same broadcast domain (LAN)
- Operates directly over link-layer technologies such as Ethernet
- Essential for delivering frames to the correct device

3) Explain ICMP protocol in detail

Ans:

Type(8 bit)	Code(8 bit)	Check Sum(16 bit)
Extended Header(32 bit)		
Data/Payload(Variable Length)		

- Internet Control Message Protocol (ICMP) works in the network layer of the OSI model and the internet layer of the TCP/IP model. ● It is used to send control messages to network devices and hosts. Routers and other network devices monitor the operation of the network.
- When an error occurs, these devices send a message using ICMP. Messages that can be sent include "destination unreachable", "time exceeded", and "echo requests".

Fields Explanation:

- **Type:** Identifies the type of ICMP message.
- **Code:** Provides additional information about the

message type.

- **Checksum:** Used for error checking of the ICMP

message.

- **Rest of Header:** Contains extra information depending on the type of message.

- **Data:** Contains the original IP packet or additional information.

Example of ICMP Usage

When we run the **ping command**, the computer sends an **ICMP Echo Request** to the destination.

If the destination is reachable, it sends back an **ICMP**

Echo Reply. Example command:

```
ping 192.168.1.1
```

UNIT 5

2 Marks

1) Define Network topology.State types of topology

Ans:

A network topology is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers, and other devices are connected and how data flows between them. **Types of topology**

- 1.Star topology
- 2.Mesh topology
- 3.Tree topology
- 4.Hybrid topology

2) List any four networking devices

Ans:

- 1.Switch
- 2.Router
- 3.Gateway
- 4.Bridge

4 Marks

1) Compare client-server and peer-to-peer model

Ans:

Client-Server Network	Peer-to-Peer Network
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In client server network, clients and servers are differentiated, Specific server and clients are present	In peer to peer network, clients and servers are not differentiated.
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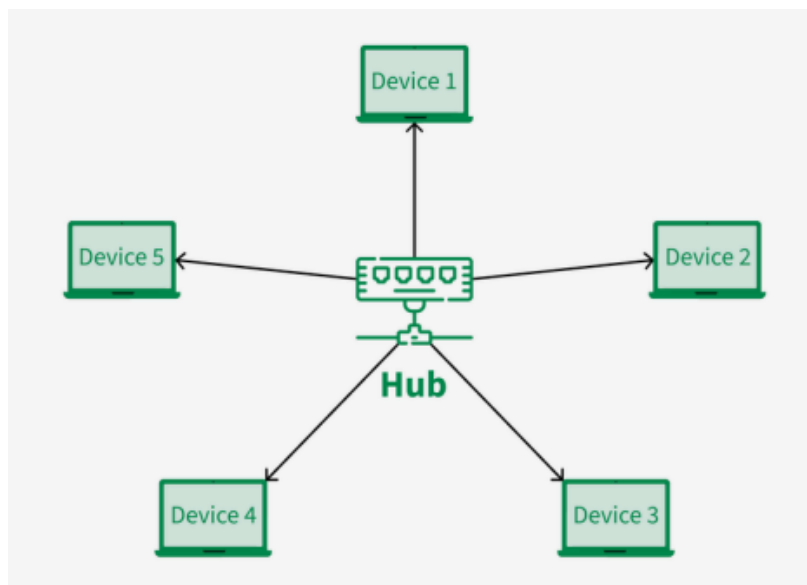
Client-Server Network focuses on information sharing.	While Peer-to-Peer Network focuses on connectivity.
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□

In Client-Server Network, Centralized server is used to store the data.	While in Peer-to-Peer Network, Each peer has its own data.
In Client-Server Network, Server respond the services which is request by Client.	While in Peer-to-Peer Network, Each and every node can do both request and respond for the services.

2) Explain Star topology with diagram.

Ans.



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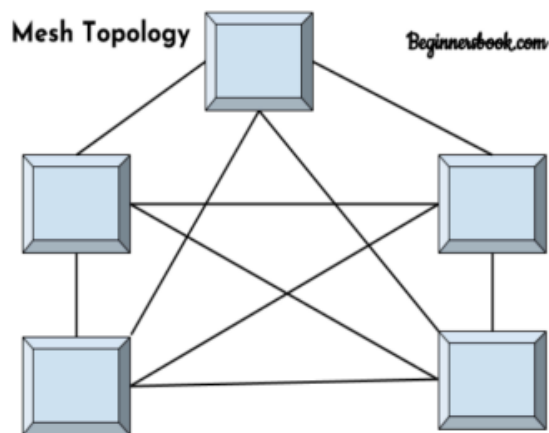
- Star Topology is a network setup in which each device is connected to a central node called a hub.
- The hub manages the data flow between the devices.
- If one device wants to send data to another device, it has to first send the information to the hub, and then the hub transmits that data to the required device.
- A star may be a Local Area Network (LAN) topology where all nodes are individually connected to a central connection point (hub).
- The number of links required to connect nodes in the star topology is N where N is the number of nodes.
- The hub or switch manages and controls all functions of the network.

6 Marks

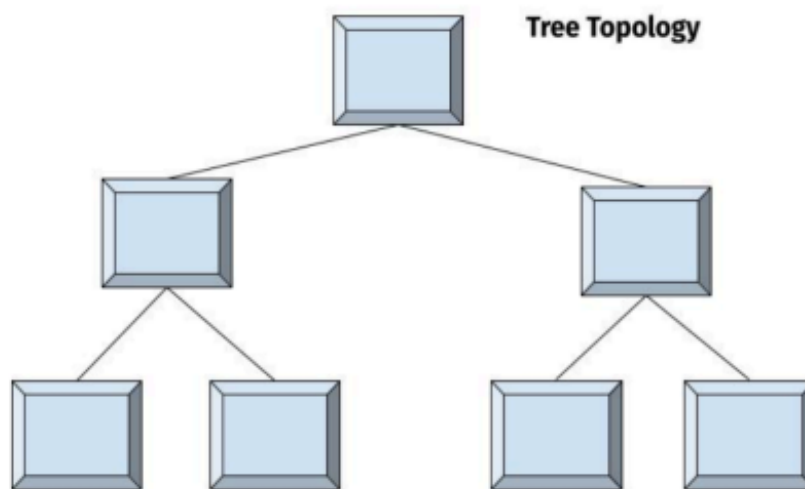
1) Explain Mesh, Tree and Hybrid topology with suitable diagram Ans:

Mesh

- In mesh topology each device is connected to every other device on the network through a dedicated point-to-point link.
- When we say dedicated it means that the link only carries data for the two connected devices only.
- Lets say we have n devices in the network then each device must be connected with $(n-1)$ devices of the network. Number of links in a mesh topology of n devices would be $n(n-1)/2$.

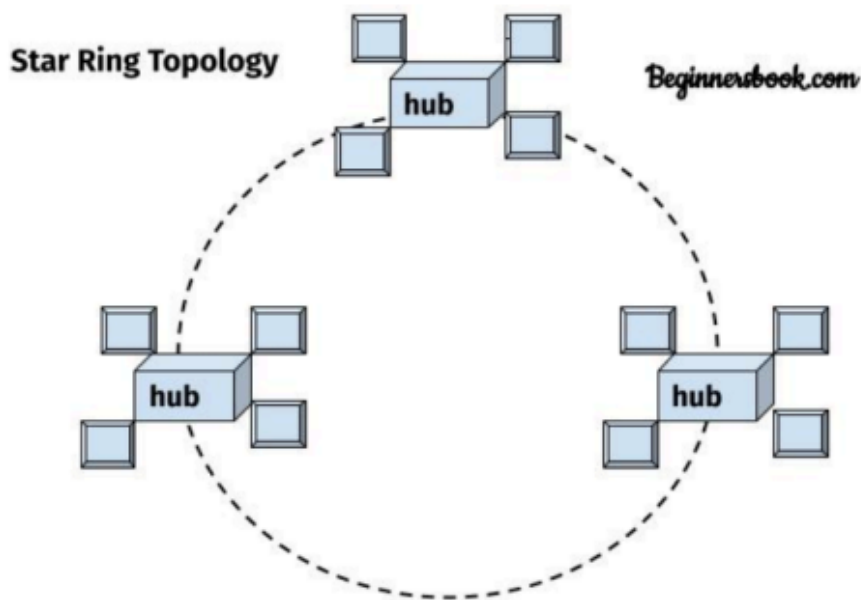


Tree



- Tree topology is an example of hybrid topology as it is a combination of star topology and bus topology that forms a tree like structure.
 - The devices are connected to each other in hierarchical manner. Each node has arbitrary number of child nodes. The device at the top of the hierarchy is known as root node and other devices that are lower in hierarchy known as descendants of the root node.
 - There is a single path between two devices and this is the path of the communication between devices.
 - Tree topology is also called star-bus topology.

Hybrid



A combination of two or more topology is known as hybrid topology. For example a combination of star and bus topology is known as star bus hybrid topology, this topology is shown in the above diagram.

- Formed by combining different topologies
- Customized according to network needs
- Common in large enterprise environments
- Supports complex network structures

2) Explain the working of Switch, Gateway and Bridge

Ans:

1. Switch

A **switch** is a networking device used to **connect multiple devices in a Local Area Network (LAN)** and forward data to the correct device.

Working of Switch:

1. When a device sends a data frame, it reaches the switch.
2. The switch reads the **MAC address** in the frame.
3. It checks its **MAC address table** to find the destination device.
4. The switch sends the frame **only to the specific port** where the destination device is connected.

Example:

In a computer lab, a switch connects many computers and sends data only to the intended computer.

2. Gateway

A **gateway** is a network device that **connects two different networks that use different communication protocols**.

Working of Gateway:

1. Data is received from one network.
2. The gateway **translates the data format or protocol**.
3. It then forwards the converted data to the other network.

Example:

A gateway can connect a **LAN network to the Internet** or connect networks using different protocols.

3. Bridge

A **bridge** is a networking device used to **connect two or more LAN segments** and reduce network traffic.

Working of Bridge:

1. The bridge receives a data frame from one LAN segment.
2. It checks the **MAC address of the destination device**.
3. If the destination is in the same segment, it **does not forward the frame**.
4. If the destination is in another segment, it **forwards the frame to that segment**.

Example:

If two office networks are connected using a bridge, it helps control unnecessary traffic between them.

